

1.1 Add Product

Use Case “Add Product”

1. Use case code

UC003

2. Brief Description

This use case describes the interaction between the Product Manager and AIMS when adding a new product to the store

3. Actors

3.1. Product Manager

4. Preconditions

1. The Product Manager must be logged in.
2. The Product Manager has the necessary permissions to add products.

5. Basic Flow of Events

1. The Product Manager selects the option to add a new product.
2. The AIMS software displays the product entry form.
3. The Product Manager enters product details (title, category, value, price, etc.).
4. The AIMS software validates the input fields.
5. The Product Manager submits the form.
6. The AIMS software stores the new product in the database.
7. The AIMS software confirms the successful addition of the product.
8. The system updates the product list.

6. Alternative flows

Table N-Alternative flows of events for UC Add Product

No	Location	Condition	Action	Resume location
1.	At Step 3	If any required field is missing or incorrect	• AIMS software highlights invalid fields	At Step 2

7. Input data

Table A-Input data for Adding a Product

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Title	Name of the product	Yes	Non-empty string	“Dune”

2.	Category	Product category	Yes	Selected from predefined list	Book
3.	Value	Base price of the product (before VAT)	Yes	Positive number	100,000
4.	Price	Selling price (before VAT)	Yes	Between 30%-150% of Value	120,000
5.	Barcode	Unique product identifier	Yes	Alphanumeric string	1234567890
6.	Description	Short product description	No	String	Classic novel
7.	Quantity	Number of items available in stock	Yes	Non-negative integer	50
8.	Warehouse Entry Date	Date the product was added to inventory	Yes	Valid date format (YYYY-MM-DD)	2025-02-28
9.	Dimensions	Product dimensions (L x W x H)	Yes	Numeric values in cm	20x15x5 cm
10.	Weight	Product weight	Yes	Numeric value in kg	0.5 kg

8. Output data

Table B-Output data for Adding a Product

No	Data fields	Description	Display format	Example
1.	Product name	Name of product	Text	Dune
2.	Price	Selling price (before VAT)	Currency	120,000
3.	Category	Category of product	String	'book'
4.	VAT	10% of the selling price	Currency	12,000
5.	Total price	Price including VAT	Currency	132,000
6.	Stock quantity	Available units	Integer	40

9. Postconditions

1. The newly added product is stored in the system.